Esimbi

Here are two sequences of five numerals written in Esimbi[1]. In each sequence, the ratio between each number and the next is the same. Note that this ratio may or may not be different across the two sequences.

Sequence 1

(a) mərakpə

(b) moŋi

(c) moŋioŋi

(d) buɣu nə malalə

(e) mobuɣu makələ nə mərakpə

Sequence 2

(a) kenə

(b) makələ

(c) matənə moŋi

(d) mobuɣu mɔrakpə nə matənə mərakpə

(e) mobuɣu moŋioŋi nə kenə

Assignment 1:

Find the Hindu-Arabic numerals (i.e. the numbers you usually write) that correspond to the two sequences above.

Assignment 2:

Following are three expressions in Esimbi, where the blank boxes are placeholders for the missing operators. All three expressions evaluate to the same number, 42.

(a) mobuɣu buɣu nə mobuɣu mɔrakpə nə malalə ☐ makələ

(b) mobuɣu matənə nə kenə ☐ matənə moŋi

(c) mobuɣu mɔrakpə nə kenə ☐ mərakpə

Write 42 in Esimbi numerals. Fill in the missing operators and write the equations in Hindu-Arabic numerals.

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Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

[1] Esimbi is a Tivoid language of southwestern Cameroon. It is also called Isimbi, Simpi, Age, Aage, Bogue and Mburugam.